

Element B: Documentation and Analysis of Prior Solution Attempts

At our independent boarding school we have an ongoing paper waste problem. Paper is being wasted because teachers and students sometimes print on only one side of the paper or make mistakes when using the printer or photocopier. This is a large-scale issue, over 200,000 pages have already been printed or photocopied this school year alone, and we still have three months until the end of the school year (Mr. B, personal communication, March 31, 2026). We need to find a solution to reuse the printed and photocopied paper to prevent it from being wasted and going into a landfill.

Pulping paper is the process of soaking broken-up paper in water to make new paper with the pulp of the old paper. The problems with this process are that it is time-consuming, messy, and the quality of the paper is inconsistent, and may not be as good as store-bought. Paper pulping may not be able to meet stakeholders' needs because of the time and the quantity of paper being produced. (Raj, 2024) (*Origami - How to Articles from WikiHow*, 2016)



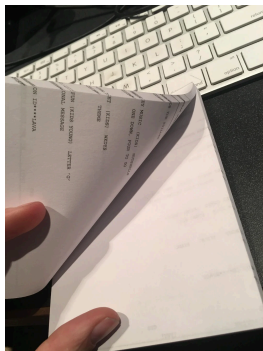
(Pulping paper)

Arts and crafts can be done to make decorations or to pass time. They can also help improve hand eye coordination and fine motor skills. Arts and crafts are less than \$2, can be made into things smaller than 8.5 x 11 inches and there are no sharp edges. The problem with making arts and crafts is that it is only temporary, and does not use large quantities of paper. (Kaintal, 2021)



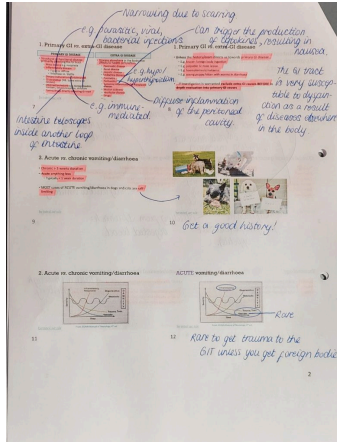
(Arts and Crafts)

Used or old paper can be used for taking notes for work or school. This method is virtually free, and it reduces unnecessary waste. Reusing paper reduces greenhouse gas emissions, and saves water, energy, and money. Only works if one side of the paper is blank, and can be disorganized and inconvenient. 90% of used paper cannot be used for notetaking, and gathering 20 pages or more of scrap paper can prove difficult. (*Origami - How to Articles from WikiHow*, 2016)



(Note pad made of used paper)

The current solution of reusing paper is using old paper for note-taking. The positives of taking notes on already used paper are that it is very cheap and inexpensive, but it can be disorganized and inconvenient, and it can be difficult to find paper with only one side blank.



(Scrap paper used as notebook paper)

References

Day, R. (2024, July 19). *How to Make Paper Pulp*. WikiHow.

<https://www.wikihow.com/Make-Paper-Pulp>

Kaintal. (2021, December 22). *6 Important Benefits of Arts and Crafts for Students - Kaintal School*. Kaintal School -.

<https://kaintalschool.com/6-important-benefits-of-arts-and-crafts-for-students/>

Raj, N. (2024, June 21). *Ways to Reuse Paper at Home*. Rescript.

<https://www.rescript.in/blogs/news/ways-to-reuse-paper-at-home>